

Development of a novel anchorage system for prestressing FRP grids

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ARTICLE INFO

Keywords:

FRP grid
Prestressing
Anchorage
Tensioning capacity
Strengthening

ABSTRACT

As an emerging strengthening technique, the use of fiber-reinforced polymer (FRP) grid and inorganic bonding materials can overcome the issues of insufficient durability and fatigue failure associated with conventional techniques, making it suitable for the strengthening of concrete structures in harsh environments. However, most current research focuses on the strengthening with non-prestressed FRP grids, which cannot achieve crack closure and deflection recovery. Due to the unique structure of the FRP grid, existing anchorages cannot effectively apply high-level prestress. To address this issue, a novel anchorage system for prestressing basalt fiber-reinforced polymer (BFRP) grids was introduced and investigated in this study. First, a finite element (FE) model was established to simulate the stress characteristics of the existing FRP grid anchorage, with the purpose of evaluating how load eccentricity, bolt pre-tightening force, and anchor plate structure affect anchorage effectiveness. Based on this, a novel anchorage was developed, taking into account aspects such as avoiding tension eccentricity, optimizing the arrangement of pre-tightening bolts, and slotting anchor plates. Furthermore, FE analysis and experiments were conducted to investigate the tensioning capacity of the proposed anchorage. The results indicate that the proposed anchorage can significantly enhance anchorage performance by effectively reducing stress concentrations in BFRP grids and improving stress coordination among longitudinal grid bars.

1. Introduction

Fiber-reinforced polymer (FRP) grid strengthening is a new technique that has been developed recently to strengthen concrete structures in harsh environments. It provides benefits such as corrosion and fatigue resistance, which can help to address the issue of poor long-term durability associated with conventional strengthening techniques [1–4]. The method involves fixing FRP grid to the surface of existing concrete structures using anchoring devices such as rivets, and painting or high-pressure spraying inorganic bonding materials to form an integral structure with the existing concrete members. Compared to traditional externally bonded FRP laminates or steel plates strengthening methods, the grid-like characteristics of FRP grid enable better adhesion with inorganic material matrices and exhibit excellent fatigue resistance. Common inorganic bonding materials include polymer cement mortar (PCM), engineered cementitious composite (ECC), and geopolymers mortar, which demonstrate good environmental adaptability and ensure the long-term durability of the strengthening system [5–7].

Researchers have used FRP grid to strengthen concrete beams [5,6,8,9], slabs [10], columns [11–13], and tunnel linings [14,15], studying

their mechanical properties and design methodologies. According to the research findings, FRP grid strengthening can effectively control the development of cracks while improving the stiffness and load-bearing capacity of concrete members. Furthermore, FRP grid strengthening technique has demonstrated positive impacts on concrete members' fatigue performance and high-temperature resistance [16–18]. However, directly bonding strengthening materials cannot achieve crack closure and deflection recovery in existing concrete structures. For concrete members with high reinforcement ratios, the enhancement of flexural stiffness is relatively limited, resulting in inefficient strengthening [19,20]. Li et al. [21], He et al. [22], and Zhou et al. [23] utilized prestressed basalt fiber reinforced polymer (BFRP) grid to strengthen concrete hollow slabs, as well as flexurally and shearly strengthen concrete beams. The experimental results demonstrated that the prestressed FRP grid strengthening technique can increase the cracking load and flexural stiffness of concrete members while effectively suppress crack propagation.

When using the prestressed strengthening technique in practice, one of the concerns is the anchorage system. An excellent anchorage system should provide a reliable anchorage effect and easy operation process at

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<https://doi.org/10.1016/j.istruc.2025.109958>

Received 4 April 2025; Received in revised form 5 August 2025; Accepted 11 August 2025

Available online 15 August 2025

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the construction site. To date, different anchorage methods have been proposed for prestressed strengthening techniques with various FRP types [24–26], including carbon fiber sheets, carbon FRP (CFRP) plates and basalt FRP (BFRP) plates. The developed FRP anchorage methods mainly include the steel plate anchorage [25,27], nonmetallic anchorage [28,29], gradient anchorage [30,31], wedge-clamp anchorage [32,33], and flat-clamp anchorage [34,35]. Existing studies have proven that the clamp anchors can effectively anchor FRP sheets/plates, and the rupture failure of FRP can even be achieved.

In the context of prestressed strengthening utilizing FRP grids, the distinctive structural properties of these composite materials render them particularly sensitive to the application of prestressing forces [21, 22]. It is difficult to ensure the coordinated stress distribution among the grid bars during the prestress tensioning process. Compared with the aforementioned anchorage methods for prestressed FRP sheets/plates, the existing FRP grid anchorages have lower anchorage efficiency and are unable to apply higher prestressing levels, thereby affecting the active strengthening effect of the concrete members.

The above analysis indicates that existing research primarily focuses on the non-prestressed FRP grid strengthening and some limited prestressed FRP grid strengthening of concrete members, but lacks in-depth investigation into effective anchorage methods for FRP grid. This study conducts a finite element (FE) analysis to investigate the main causes of FRP grid premature failure during the prestress tensioning process. Based on this, a novel FRP grid anchorage system is designed, and FE analysis is used to examine its mechanical properties. Furthermore, the effectiveness of this anchorage system in improving the anchorage efficiency of the BFRP grid is validated through tensioning tests.

2. Analysis of the existing FRP grid anchorage

The existing research has developed several typical prestressed FRP grid anchorages, but they exhibit issues of low tensile anchorage efficiency and premature fracture of FRP grids. This study aims to propose a novel anchorage system for prestressing FRP grids based on the existing anchorages to achieve higher tensile anchorage efficiency, thereby promoting the application of prestressed FRP grid strengthening technology.

2.1. Existing FRP grid anchorages

There is currently very little research on prestressed FRP grid strengthening. Li et al. [21] created a dispersed anchorage system for prestressing BFRP grid bars successively, as shown in Fig. 1(a). Each grid bar was sequentially tensioned using a waveform anchor, which was then screwed onto the angle steel. Anchor bolts were used to fasten the angle steel to the concrete surface. This method has low tension force for each grid bar, making it convenient for construction. However, the sequential tensioning of grid bars can greatly impact their coordinated stresses, potentially lead to excessive deformation of the transverse grid

bars and premature of the FRP grid. He et al. [22] developed an integral anchorage system using steel plate anchorages and fastening bolts to anchor and tension the BFRP grid, as shown in Fig. 1(b). This technique applies uniform force to each grid bar simultaneously during prestress tensioning. However, stress imbalance and localized fractures of FRP grids are still observed during testing. Moreover, the FRP grid experiences additional stress as a result of the noticeable tension eccentricity produced by this integral anchorage design.

2.2. FE model

To improve the overall prestress tensioning performance of the FRP grid, this study chose the aforementioned integral anchorage (Fig. 1(b)) for further investigation. The following sections used the general FE software ABAQUS to simulate the tensioning load characteristics of the existing integral anchorage and to analyze the factors that may cause the FRP grid to fail prematurely.

The FE model was established based on the integral anchorage system of BFRP grids described in reference [22]. The model consisted of upper and lower anchor plates, bolts, BFRP grids, and a threaded steel rod, as shown in Fig. 2. The BFRP grid measured a spacing of 50 mm × 50 mm, with a grid bar width of 10 mm and a thickness of 3 mm. The threaded steel rod is located 50 mm from the anchor plate. Each component of the anchorage system and the BFRP grid were modeled using eight-node elements (C3D8R). The interaction between the anchor plate and the BFRP grid was simulated utilizing surface-to-surface contact. The tangential direction employed a penalty function with Coulomb friction, with the tangential friction coefficient set to a constant value of 0.26 [36]. In addition, the normal contact was set as hard contact. The loading process included two stages: bolt pre-tightening and prestress tensioning. During the bolt pre-tightening stage, bolts load with 10 kN was adopted to apply pre-tensioning force to the BFRP grid. During the prestress tensioning stage, a 10 mm displacement was applied to the end of the threaded steel rod for tensioning.

The threaded steel rod, pre-tightening bolts, and anchor plates were considered as ideal elastic-plastic materials with the following parameters: yield strength of 235 MPa, elastic modulus of 200 GPa, and Poisson's ratio of 0.3. Table 1 describes the material parameters of the BFRP grid, which was set as an anisotropic linear elastic material with references to the literature [22]. The BFRP grid is divided into three regions: longitudinal bar, transverse bar and grid node for material assignment. In the table, E_1 , E_2 , and E_3 represent the elastic modulus of the BFRP grid bars in x, y and z directions, respectively. The mechanical properties of the grid node in both longitudinal and transverse directions are characterized by the respective directional properties of the longitudinal fibers, i.e., both E_1 and E_2 are set as 21 GPa, as shown in Table 1.

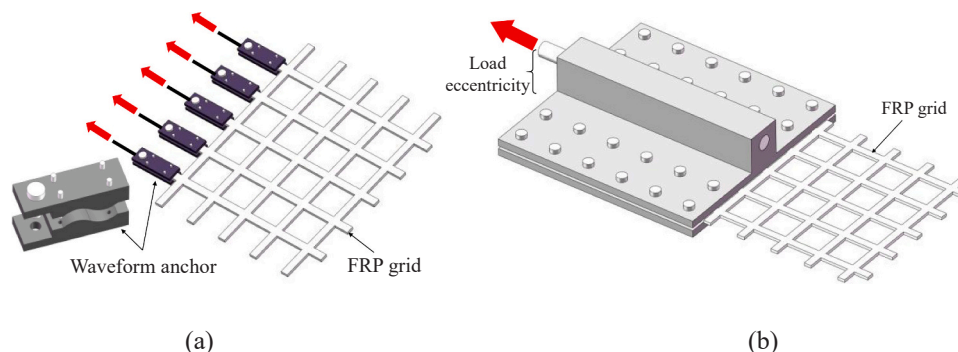


Fig. 1. The existing FRP grid anchorages: (a) dispersed anchorage [21]; (b) integral anchorage [22].

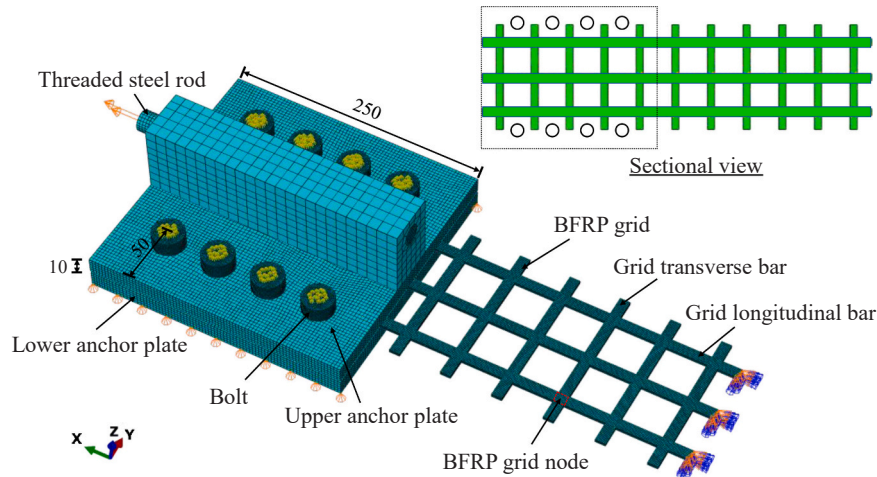


Fig. 2. FE model of the existing BFRP grid anchorage (mm).

Table 1

Material properties of the BFRP grid adopted in the FE model.

Component parts	Elastic modulus (GPa)			Shear modulus (GPa)	Poisson ratio
	E_1	E_2	E_3		
Longitudinal bar	21	6	6	6	0.25
Transverse bar	6	21	6	6	0.25
Grid node	21	21	6	6	0.25

2.3. Deformation and stress analysis

Due to the existence of tension eccentricity, the BFRP grid gradually warped as load increased, as shown in Fig. 3. By analyzing the clamping pressure distribution of the BFRP grid (Fig. 4), it is found that the pressure at the anchorage end decreased after tensioning. This indicates that the tension eccentricity not only affects the overall clamping effect of the BFRP grid, but also exacerbates the slip problem at the anchorage end. Since the bolts were only distributed on the outsides of the BFRP grid, the center of the anchor plates arched after tensioning, providing ineffective clamp for the inner grid bars.

Fig. 5 illustrates the axial stress distribution of the BFRP grid during the tensioning phase. The axial stress gradually decreased from the anchorage end to zero, with this distance defined as the stress transfer length of the anchorage. During this process, the tensioning load was gradually transferred from the BFRP grid to the anchor plates. Fig. 5(a) indicates that the stress transfer length of the inside longitudinal bar is significantly longer than that of the outside longitudinal bar. This is because the existing anchorage is only provided by the outer single row of bolts, and the inner longitudinal rod is farther away from the bolt than the outer longitudinal rod, resulting in no effective clamping. As a result, it is prone to premature failure, which occurs when a single longitudinal bar fails while the other longitudinal bars have not achieved their ultimate condition. To further investigate the difference of each longitudinal bar, the axial stress distribution across the cross-sections was plotted,

as shown in Fig. 5(b). In addition to the uneven stress distribution between longitudinal bars, the outside longitudinal bars also exhibited an uneven stress distribution. This is because only one row of bolts clamped the outside longitudinal bars, and the clamping effect gradually diminished outside from the bolts.

According to previous test results [21,22], tensile failure of the BFRP grid occurs mostly at the grid nodes, where shear stress is introduced by the action of the transverse bar. According to the Hashin's criterion for composite materials [37], the presence of shear stress could lead to a reduction in the longitudinal tensile strength of the FRP composites (Eq. (1)), resulting in premature failure.

$$\left(\frac{\sigma_{11}}{X_T}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2 \geq 1 \quad (1)$$

where σ_{ij} ($i, j = 1, 2, 3$), X_T and S_{ij} ($i, j = 1, 2, 3$, $i \neq j$) represent the magnitude of the stress, tensile strength and shear strength, respectively.

The shear stress distribution on the BFRP grid after tensioning was plotted in Fig. 6. The transverse bars on both sides of the BFRP grid exhibited a notable vertical shear stress (S_{13}), as seen in Fig. 6(a). This is attributed to the deformation of the anchor plate. Despite the warping of BFRP grids, there was no obvious vertical shear stress produced close to the anchorage end. This is probably because the BFRP grid can freely deflect at the anchorage end due to the anchor plate's arching. As shown in Fig. 6(b), the transverse shear stress (S_{12}) was mainly occurred at the first row of grid nodes from the anchorage end. This is attributed to the fact that transverse bars limit the deformation of longitudinal bars within the stress transfer length. In comparing to Fig. 4(b), it is found that the transverse shear stress increased as the corresponding clamping pressure grew.

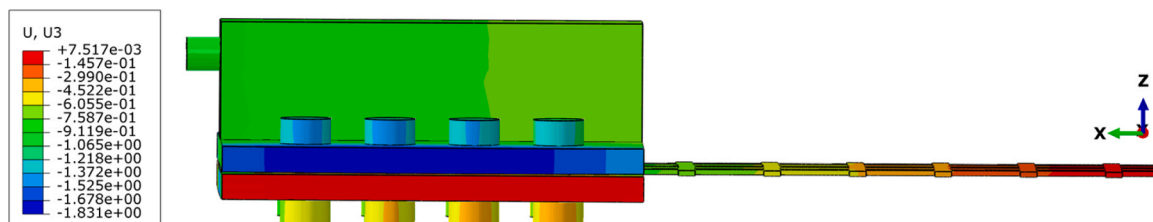


Fig. 3. Vertical deformation of the existing anchorage after tensioning (mm).

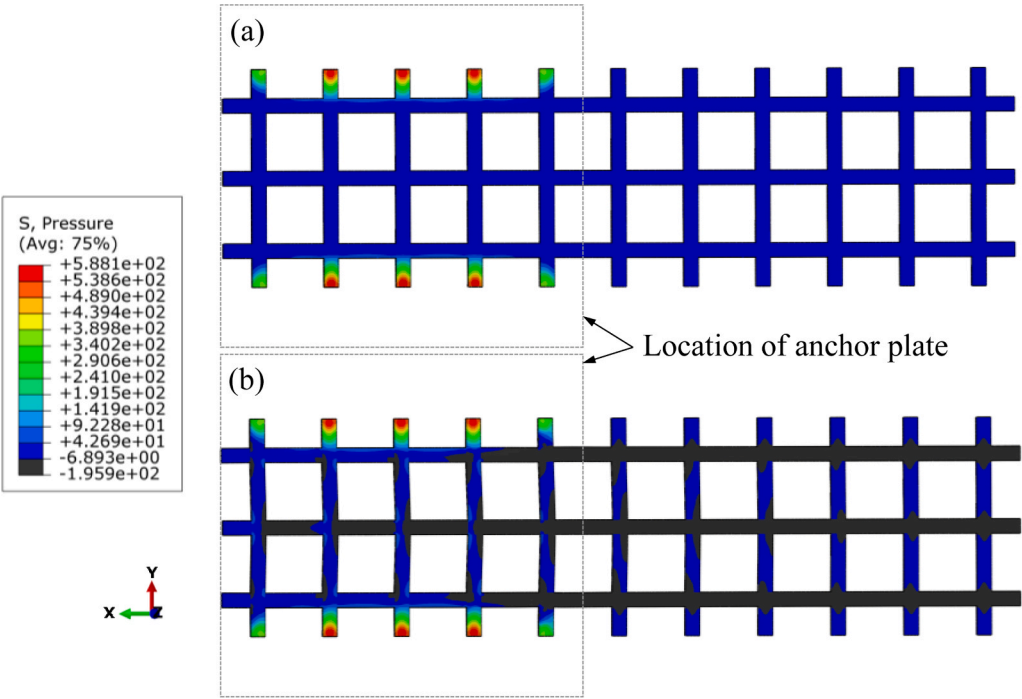


Fig. 4. Clamping pressure distribution of the BFRP grid with the existing anchorage (MPa): (a) pre-tightening stage; (b) tensioning stage.

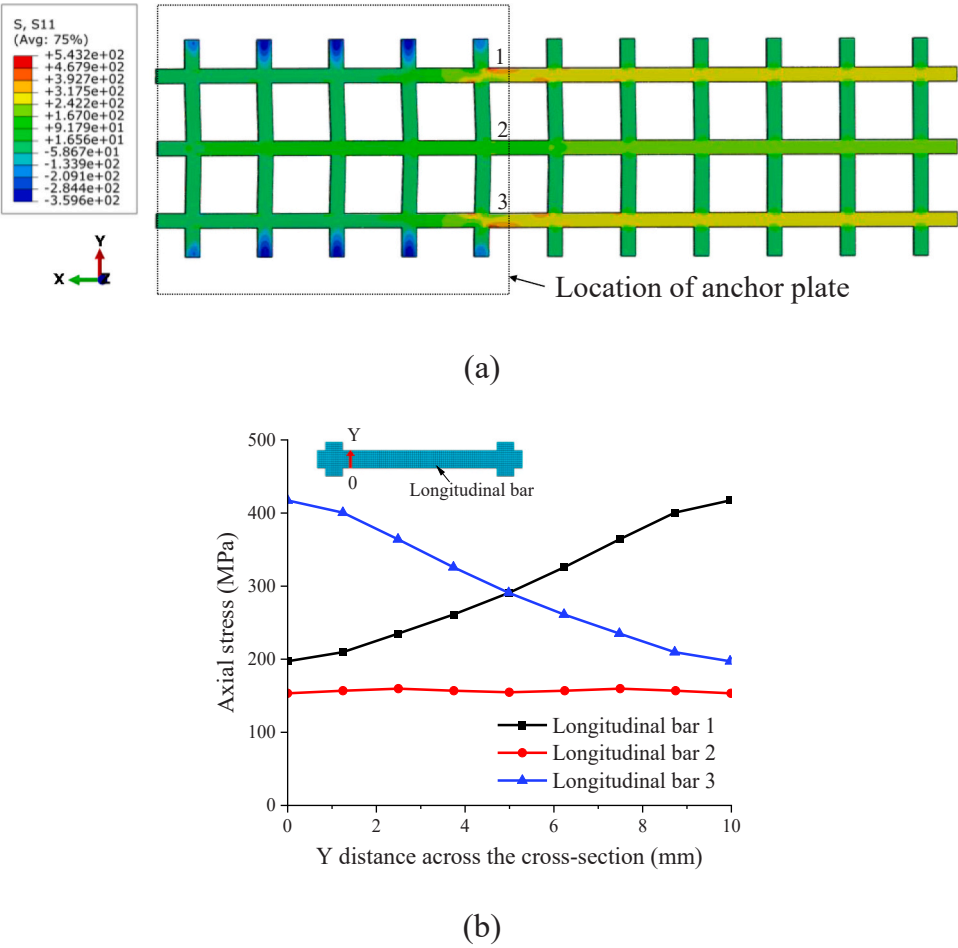


Fig. 5. Axial stress of the BFRP grid with the existing anchorage: (a) stress distribution (MPa); (b) axial stress across the cross-section.

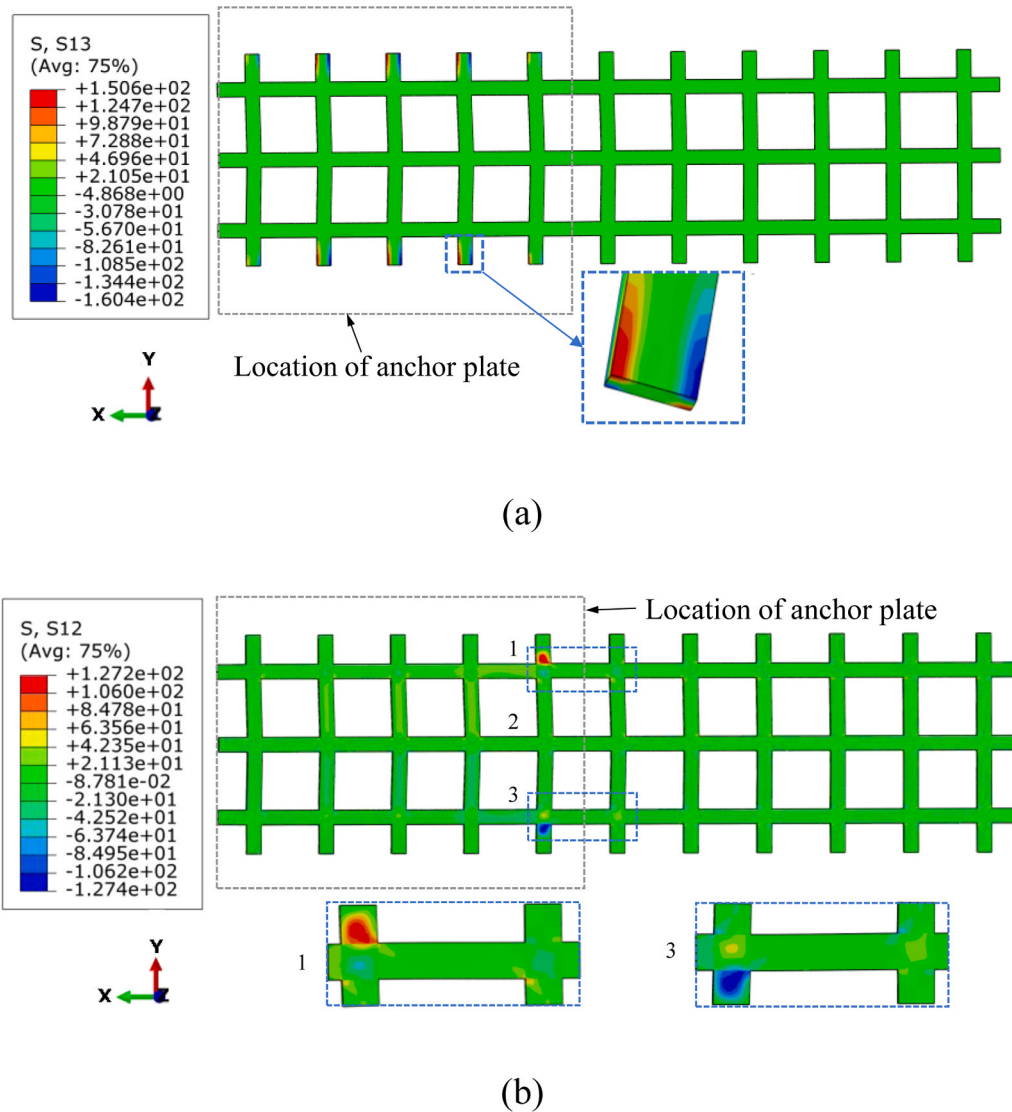


Fig. 6. Shear stress distribution of the BFRP grid with the existing anchorage (MPa): (a) vertical shear stress; (b) transverse shear stress.

3. Proposing of a novel anchorage system

3.1. Design scheme

Based on the aforementioned analysis of the existing anchorage, a novel anchorage system for the prestressing of BFRP grids was proposed, as shown in Fig. 7. The main improvements are as follows:

- (1) Threaded steel rods were positioned on both sides of the anchor plate to mitigate the load eccentricity. A front guard plate was installed at the anchorage end to ensure that the threaded steel rod's tensioning force was evenly transferred to the anchor plate.
- (2) To address the uneven stress distribution between the inner and outside longitudinal bars, bolts were added to the inner side of the anchor plate to ensure that all longitudinal bars were clamped by double-row bolts, thereby optimizing the stress coordination of the longitudinal bars. The threaded steel rod also limited vertical deformation on both sides of the anchor plate, resulting in a more uniform stress distribution on the longitudinal bars.
- (3) The upper anchor plate at the first row of transverse bars was slotted to alleviate the severe shear stress at the first row of grid nodes. Shear stress concentration was adequately addressed by this approach since the anchor plate had received the entire load of the second row of transverse bars.
- (4) By reducing the distance between the first row of bolts and the anchorage end, the clamping force at the anchorage end could be significantly increased, improving the anchorage system's overall stability and safety.

In addition, it is noteworthy that the removal of transverse bars also represents a potential measure to reduce the shear stress and enhance

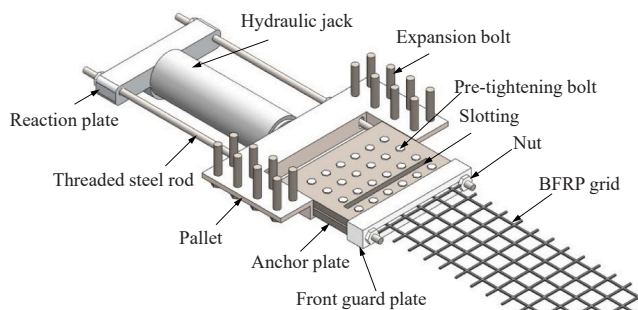


Fig. 7. The proposed anchorage system for the prestressing of BFRP grids.

the mechanical performance of grid nodes. The authors previously conducted tests in which the transverse bars in the first row of the grid were removed. The measured maximum tensile force was slightly lower compared to the case without removal. This indicates that cutting the transverse bars does not effectively enhance anchorage efficiency. The primary reason for this phenomenon is that even after removing the transverse bars, transverse fibers remain at the grid nodal points. When the grid is subjected to tensile force, the longitudinal bars still experience constraint-induced shear stress, thereby reducing their tensile strength. Furthermore, the transverse bars can assist the longitudinal bars in bearing partial axial forces. Consequently, removing the transverse bars may instead reduce the overall tensile performance of the FRP grids. Consequently, this study does not adopt the aforementioned measure.

3.2. FE analysis of the proposed anchorage system

In this section, the FE analysis method was employed to model the proposed anchorage system, aiming to preliminarily verify its feasibility. As illustrated in Fig. 8, model A was used to assess the impact of threaded steel rod position changes and bolt distribution on the structure's stress distribution, whereas model B focused on validating the improvement effects of slotting the anchor plate and shortening the length of the anchor plate ends on the shear stress of the grid node. The same materials and load steps were adopted in these two models as that presented in Section 2.2. Dimensions of anchor plates and front guard plate were illustrated in Fig. 8. The pre-tightening bolts were grade 8.8

M16 hexagonal head bolts, with grade 8 M16 hexagonal nuts. The BFRP grid bars were tensioned using displacement control, with a 10 mm displacement applied to both sides of the front guard plate, as shown in Fig. 8.

Fig. 9 illustrates the axial stress distribution of the proposed anchorage systems. As shown in Fig. 9(a), the use of double-row bolts resulted in a more uniform stress transfer length for the inside and outside longitudinal grid bars, thereby reducing the possibility of premature failure of a single bar. As illustrated in Fig. 9(b), slotting and shortening the anchor plate increased the clamping force at the anchorage end, greatly reducing stress transfer length and stress at the first row of grid nodes.

Fig. 10 illustrates the transverse shear stress (S12) distribution for the proposed anchorage system. Model A optimization resulted in a more uniform shear stress distribution between the inner and outer grid nodes, as shown in Fig. 10(a). However, significant shear stresses remained. As shown in Fig. 10(b), the loads transmitted to the grid nodes are smaller due to the improved clamping effect at the anchorage ends, lowering the possibility of premature failure. Additionally, the slotting of anchor plates alleviated the adverse stress condition of the first row of transverse bars. The stress state of the grid nodes inside the anchorage is similar to that outside the anchorage, thus further reducing the possibility of premature failure.

Fig. 11 illustrates the distribution of vertical shear stress (S13) for the proposed anchorage system. There remained a certain level of vertical shear stress at the edge of the transverse bars in Model A, as shown in Fig. 11(a). This is mainly attributed to the anchorage deformation under

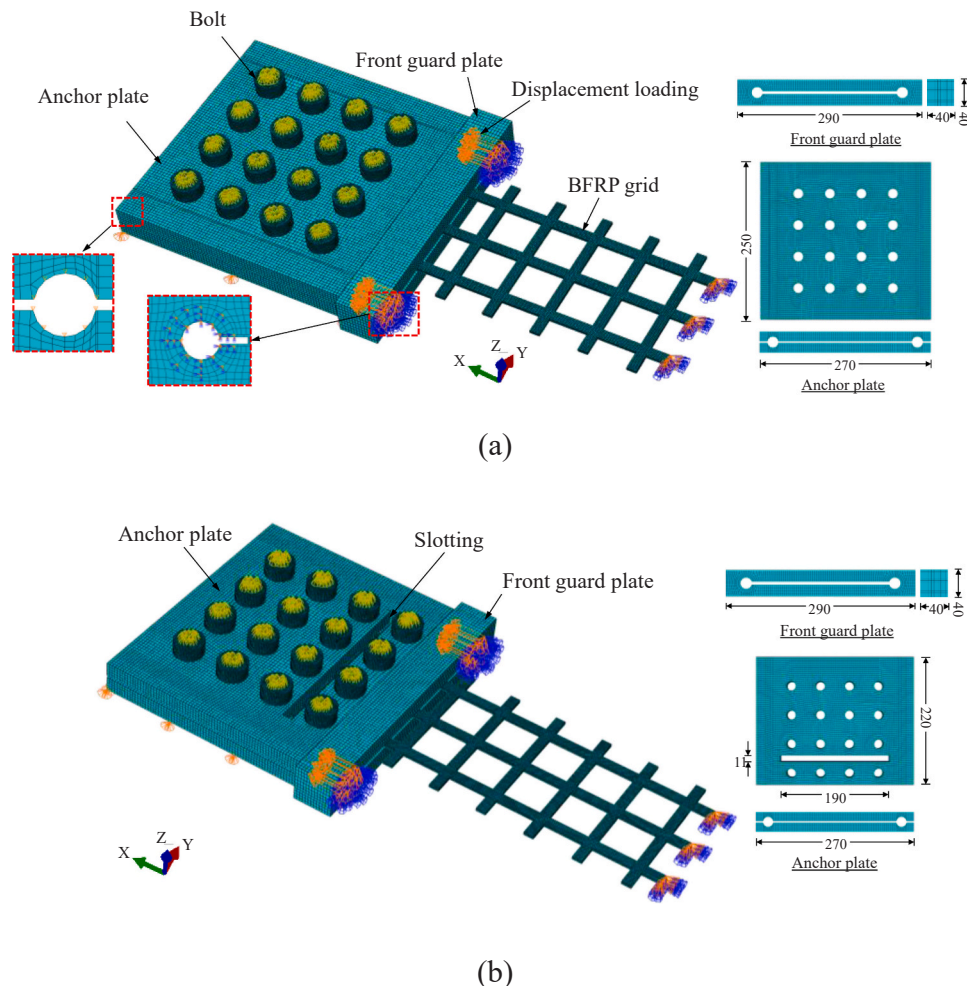


Fig. 8. FE model of the proposed anchorage system: (a) model A; (b) model B.

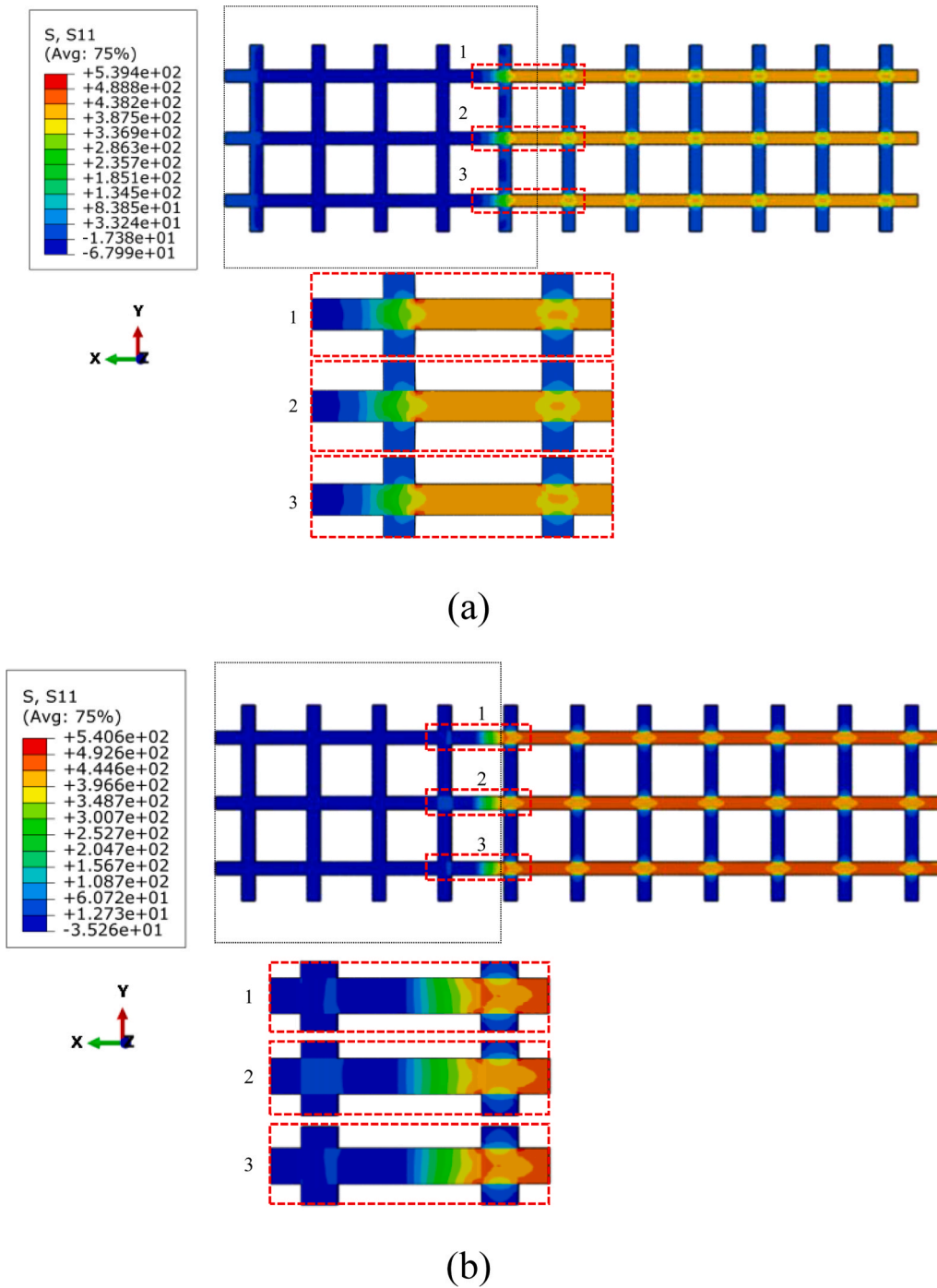


Fig. 9. Axial stress distribution of the proposed anchorage: (a) model A; (b) model B.

the pretension force. This can be mitigated by reducing the pretension force or employing anchor plates with larger stiffness to decrease the vertical shear stress. As shown in Fig. 11(b), the longitudinal bars at the anchorage end experienced inverted vertical shear stress, owing to the high pretension force of the end bolts. According to the analysis in Fig. 9 (b), the stress transfer length at this point remained within a safe distance of the second row of grid nodes. Therefore, reducing the pretension force of the bolts can effectively diminish the shear stress at the anchorage end.

3.3. Anchorage efficiency coefficient

To quantitatively assess the anchorage performance, the anchorage

efficiency coefficient η is often used in standards through anchorage tension tests (Eq. (2)) [38]. As no failure criterion was considered, the aforementioned FE simulations were unable to obtain the failure load directly. Instead, this study defined the ultimate condition as the state that the maximum axial stress in the BFRP grid reached its ultimate strength. Using FE results at this ultimate condition, a new anchorage efficiency coefficient k was introduced to assess the anchorage performance in this study. According to the previous FE investigations, two major factors contributed to the premature failure of the BFRP grid with anchorages: uneven axial stress distribution among longitudinal grid bars and shear stress concentration at grid nodes. A tension reduction factor k_0 was introduced to account for the uneven stress of longitudinal bars (Eq. (3)). In addition, a shear stress reduction factor k_1 was

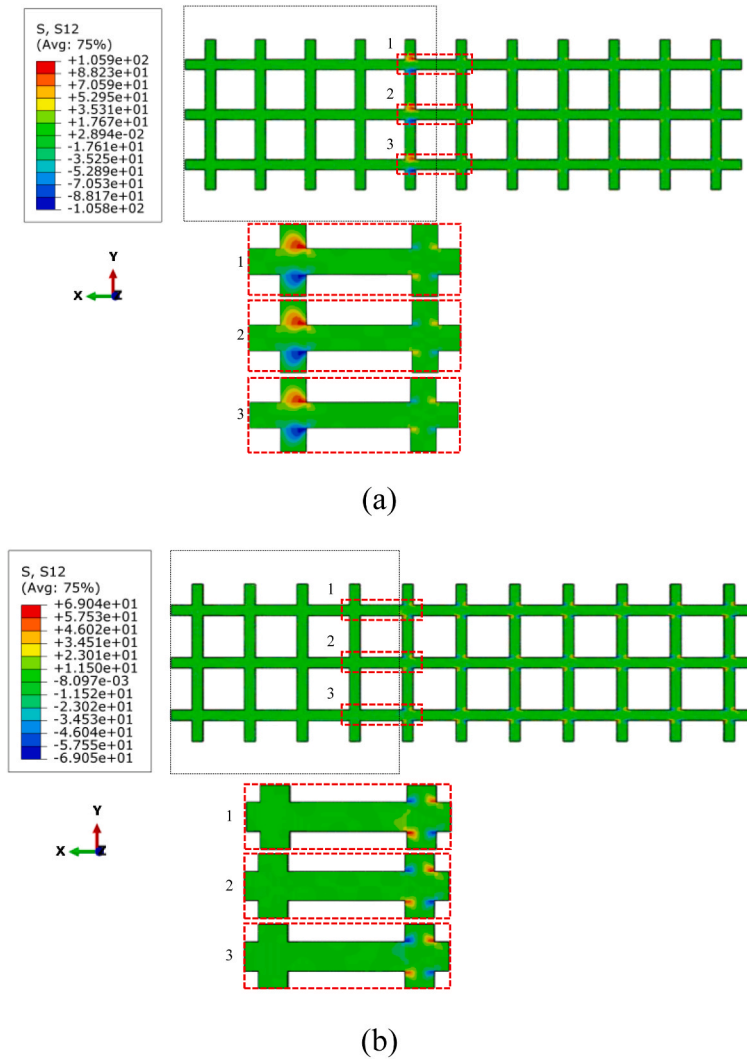


Fig. 10. Transverse shear stress distribution of the proposed anchorage: (a) model A; (b) model B.

introduced to account for shear stress concentration (Eq. (4)), which was deduced using Hashin's criterion (Eq. (1)). As a result, the introduced anchorage efficiency coefficient k can be calculated using the product of k_0 and k_1 (Eq. (5)).

$$\eta = \frac{F_{tu}}{F_{tk}} \quad (2)$$

$$k_0 = \frac{F'_{tu}}{F_{tk}} \quad (3)$$

$$k_1 = \sqrt{1 - \left(\frac{\sigma_{12}}{S_{12}}\right)^2 - \left(\frac{\sigma_{13}}{S_{13}}\right)^2} \quad (4)$$

$$k = k_0 \times k_1 \quad (5)$$

in which, F_{tu} is the maximum force obtained from static tensioning tests; F_{tk} is the ultimate load calculated according to the tensile strength of the BFRP grid; F'_{tu} is the ultimate tensile load of the BFRP grid in FE simulations; S_{12} and S_{13} are the transverse and vertical shear strengths of the BFRP grid, respectively; and σ_{12} and σ_{13} are the transverse and vertical shear stresses of the BFRP grid at the point of maximum axial stress in FE simulations, respectively.

Based on the test results in the next section and the existing study [39], the BFRP grid's tensile and shear strengths were determined to be

$X_T = 433$ MPa and $S_{12} = S_{13} = 138$ MPa, respectively. The other parameters were determined by FE simulations. As shown in Fig. 12, the anchorage efficiency coefficient k was calculated for each type of anchorage. It has been discovered that the existing anchorage may cause the premature failure of BFRP grid at a relatively low load. In comparison, the anchorage efficiency coefficient of the proposed anchorage reached up to 79.6 %, representing a 77.7 % increase over the existing anchorage.

4. Experimental validation

To validate the effectiveness of the proposed anchorage system, a model test was conducted according to the previous FE modes. A series of static tensile tests were carried out on the BFRP-anchor assembly. In this study, two kinds of anchors were used, and the same BFRP grid was used for stretching.

4.1. Test materials and method

The BFRP grid used in this experiment was manufactured by Jiangsu GMV Co., Ltd, China. The dimensions of the BFRP grid are the same as those described in Section 2.2, and the tensile properties of a single grid bar were tested using the Chinese standard GB/T 36262-2018 [40]. The tensile specimen was anchored by steel plates filled with epoxy resin, while the central 500 mm served as the test length. Each tensile

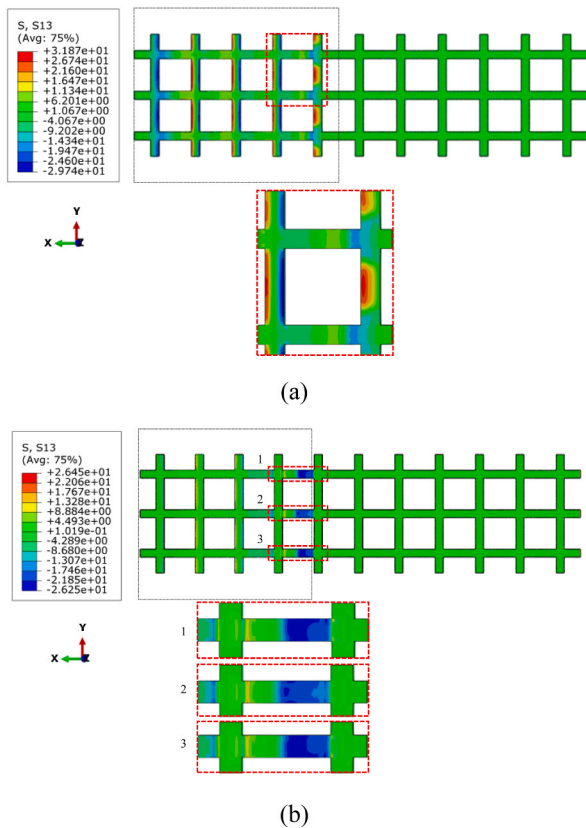


Fig. 11. Vertical shear stress distribution of the proposed anchorage: (a) model A; (b) model B.

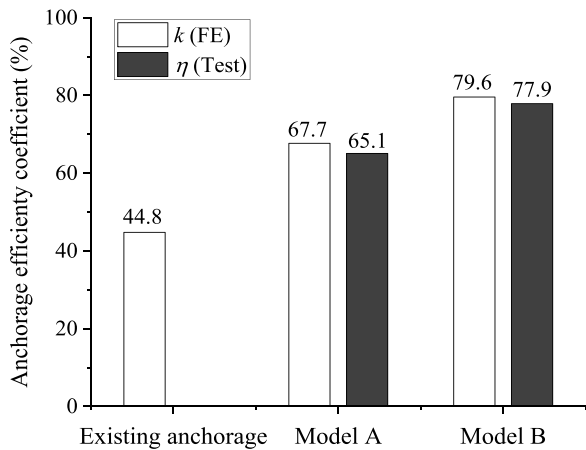


Fig. 12. Comparison of anchorage efficiency coefficient.

specimen exhibited linear elastic behavior, as illustrated in Fig. 13(a). All of the tensile specimens failed due to fiber fracture at the grid nodes, as illustrated in Fig. 13(b). Five tensile specimens were loaded to determine the average results. The measured average tensile strength, elastic modulus, and elongation values are 433 MPa, 21.1 GPa, and 2.05 %, respectively.

Two groups of anchorage specimens were designed and produced according to the FE models described in Section 3.2. The specific dimensions of the two sets of anchors are shown in Fig. 8. The test was divided into two groups: group A and group B. Group A used unslotted anchorage, group B used slotted anchorage, and BFRP grids used the same material. In order to avoid the random error of single test, each group had three duplicate specimens, as listed in Table 2. Both the

anchorage and tensioning ends of the BFRP grid were anchored using this anchorage, as shown in Fig. 14. The test length of the BFRP grid between anchorages was set to 500 mm for both anchorage specimen groups. The upper and lower anchor plates were clamped by pre-tightening bolts. The front guard plate was designed with a transverse slot to facilitate the installation of BFRP grids. Two threaded steel rods were placed on either side of the anchor plates. A digital torque wrench was used to control the bolts' pre-tightening force, which was set to 10 kN.

Test setup for the anchorage specimen is shown in Fig. 15. The testing machine clamped a back guard plate of the anchorage, applying tension force to the front guard plate via two threaded steel rods. The testing machine loaded the specimen in tension at a rate of 1 mm/min until the BFRP grid failed.

4.2. Test results and analysis

4.2.1. Failure mode

Fig. 16 shows the typical failure modes of the BFRP grid after removing the anchor plates. Both groups of specimens exhibited fiber fracture at the grid nodes, followed by splitting of the longitudinal grid bars. The longitudinal grid bar of group A failed by splitting within the anchorage, whereas group B failed by splitting outside of the anchorage. Although group A specimens reduced the eccentric tension effect, the first row of grid nodes still experienced high shear stress concentrations, leading to fiber fracture at a lower load, as shown in Fig. 16(a). In contrast, the grid node fracture location in group B specimens extended to the second row of grid nodes at a higher load. Then, the first row of the transverse bar was severely deformed, as shown in Fig. 16(b). This verified the effectiveness of slotting in releasing grid node shear stress concentrations. The stress conditions in the first row grid nodes were comparable to those outside the anchor plate.

4.2.2. Load-slip relationship

Fig. 17 presents the load-displacement curves obtained from the tensile tests, where both the load and displacement were directly measured by the testing machine. The ultimate load in group B was significantly higher than that in group A. This is because the failure occurred at the second row of nodes, which experienced a lower longitudinal stress than the first row under the same tensioning load.

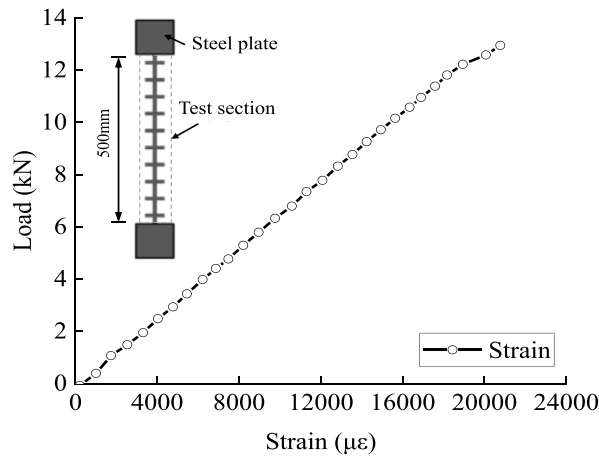
4.2.3. Anchorage efficiency

Table 2 lists the anchorage efficiency coefficient η , which was calculated using Eq. (2) based on the above test results. As illustrated in Fig. 12, the anchorage efficiency coefficients from the tests agree well with those from the FE results, demonstrating the effectiveness of the evaluation method proposed in Section 3.3.

The proposed anchorage for BFRP grids has an average anchorage efficiency coefficient of 77.9 %, which is close to that of BFRP plates (about 85 %) [26]. This suggests that the proposed anchorage can greatly increase the anchorage performance of BFRP grids while utilizing their ultimate tensile capabilities effectively. As a result, this anchorage technique has the ability to increase the prestressing level in concrete members strengthened with prestressed BFRP grids. More research is needed to study their strengthening performance under static and fatigue loading.

5. Conclusions

This study developed a novel anchorage system for the strengthening of concrete members with prestressed BFRP grids. FE analysis was employed to investigate the issues of the existing FRP grid anchorage and the superiority of the proposed one. Furthermore, the effectiveness of the proposed anchorage system was verified by the static tensile tests. Based on the investigation presented in this study, the following conclusions can be drawn:



(a)



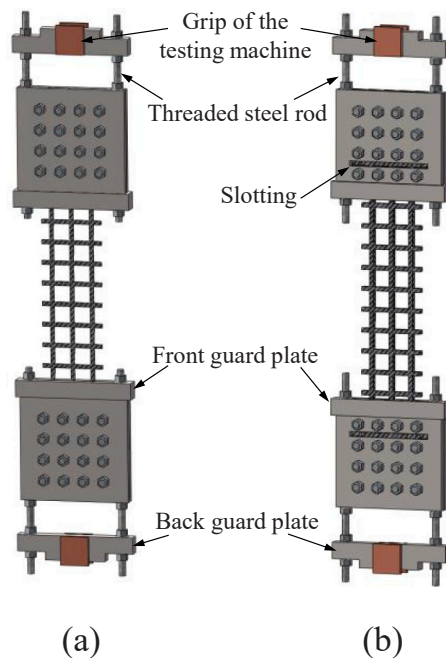
(b)

Fig. 13. Testing the material properties of the BFRP grid: (a) typical load-strain curve; (b) failure mode.

Table 2

Test results of the BFRP grid anchorage specimens.

Specimen	Maximum load F_{tu} (kN)	F_{tu}/F_{tk} (%)	Average anchorage efficiency coefficient η
A-1	24.6	63.1	65.1 %
A-2	25.8	66.2	
A-3	25.8	66.2	
B-1	30.8	79.0	77.9 %
B-2	29.9	76.7	
B-3	30.5	78.3	



(a)

(b)

Fig. 14. Schematic diagram of the anchorage specimen: (a) group A; (b) group B.

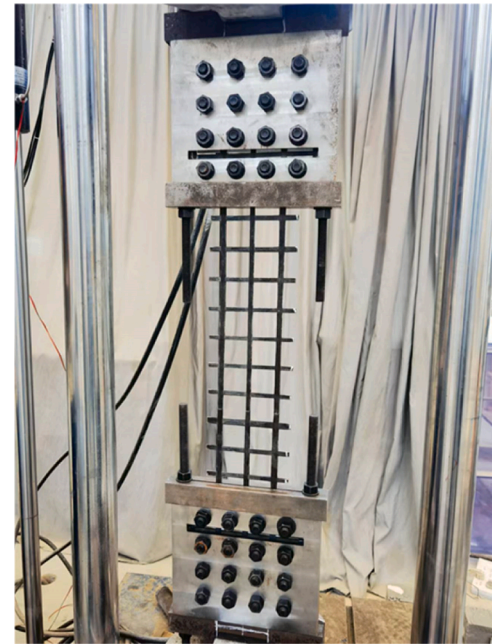


Fig. 15. Test setup of the anchorage specimen.

increased arching of the upper anchor plate. Second, insufficient bolts on the anchor plate results in uneven stress distributions throughout the longitudinal grid bar sections. Third, tensile loading causes high transverse shear stress at grid nodes within the stress transfer length of the anchor plate.

- (2) To address the concerns raised above, a novel anchorage system was developed (Fig. 7). FE analysis on the anchorage and tensioning of the BFRP grid shows that FRP grid warping and anchor plate arching are significantly reduced. The longitudinal bars in the anchor plate are well clamped, reducing the problem of uneven stress distribution. Furthermore, the shear stress concentration at the grid nodes has significantly decreased.
- (3) According to the tensioning test of the proposed anchorage system, the FE analysis results agreed well with the test results. The proposed BFRP grid anchorage achieved an average anchorage efficiency of 77.9 % during testing, which is close to that of BFRP

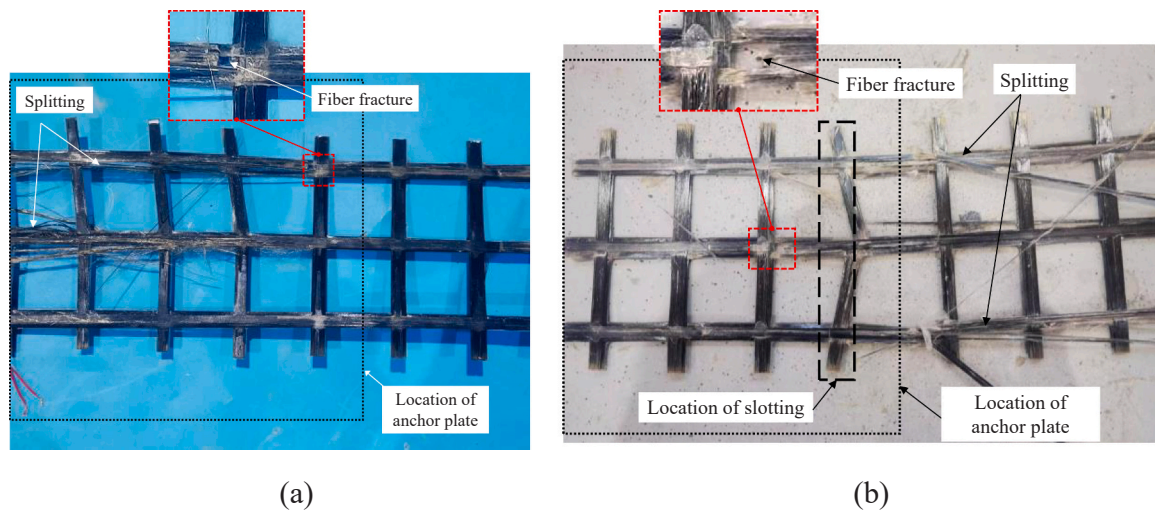


Fig. 16. Typical failure modes of the BFRP grid after removing anchor plates: (a) group A; (b) group B.

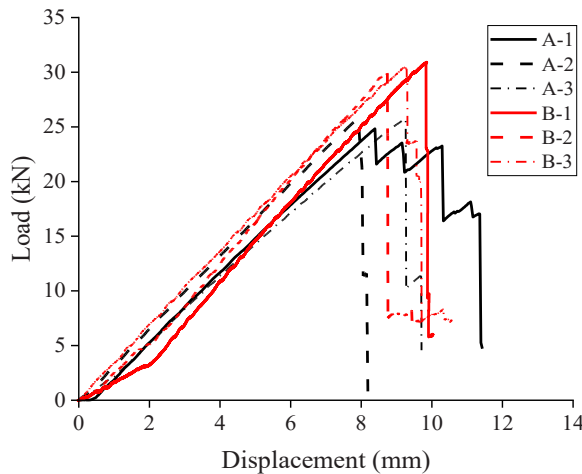


Fig. 17. Load-displacement curves of the anchorage specimens.

plates. The proposed anchorage system has been validated to effectively mitigate the problem of premature in BFRP grid and is beneficial to improving prestressing levels in concrete members strengthened with prestressed BFRP grids.

CRedit authorship contribution statement

Hao-Yu Li: Writing – original draft, Validation, Software, Methodology. **Jia-Wei Shi:** Writing – review & editing, Funding acquisition. **Yong-Sheng Tang:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the National Natural Science Foundation of China (51508157) and a grant from Engineering Research Center of Intelligent Construction and Industrialization, CAAC (MHJGKFKT-03).

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